Self-organizing reinforcement learning: a swarm approaches the explore–exploit optimum

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Abstract

The explore/exploit trade-off is the core problem of reinforcement learning (RL). We show that a minimal foraging swarm *approaches the optimum*—the ideal-free / Nash-equilibrium allocation of the multi-agent sharing game (an equilibrium, not a return maximum)—with no learner and no reward signal: self-organization does the work. Agents explore by heavy-tailed (Lévy) flights, recruit neighbours to a discovery through a threshold cascade on their proximity network, and disperse again as a patch crowds. This cascade cycle automatically balances exploration against exploitation and drives the collective toward the optimal allocation of foragers across patches—the *ideal free distribution* (foragers in proportion to richness; the Nash equilibrium of the habitat-sharing game). We are careful about what "reaches" means: the swarm attains the *shape* of this optimum (committed foragers track richness with slope 0.94) but not its exact equilibrium—it remains measurably exploitable. Because resource is shared, total intake is the same for any interior allocation, so the optimum is a load-balancing *equilibrium*, not a total-harvest maximum; the harvesting-efficiency side of foraging is analysed in the companion paper (Wilkerson, 2026). The matching is reached dynamically (within ∼150 steps) and is produced by the coupling of recruitment to movement: remove the coupling and it collapses (slope $0.94 \rightarrow 0.08$). We are deliberate about *where* the reinforcement learning lives: at the level of the *problem* and the *allocation reached*, not in the micro-dynamics. No step of the rule evaluates payoff—recruitment responds to a local crowd signal—so the mechanism is reward-free *by construction*; because under resource sharing that crowd signal is informationally tied to payoff, "reward-free" is a statement about the implementation, not a claim that no payoff-equivalent information is used. The swarm also *reaches* this allocation without *learning* it: its quality plateaus and re-settles after the resources move, with no cumulative gain. Against a trained reinforcement-learning baseline the comparison is two-sided. In an *easier, disembodied* version of the game a learner reaches the optimum more precisely. In the *same embodied* environment as the swarm—matched sensing, movement, and actions—the reward-free swarm, at zero training, reaches ≈65% of a validated oracle ceiling, while properly-trained independent-PPO and centralised-critic (CTDE) learners converge *above* it, to ≈74–75% (16 seeds each, every seed above the swarm mean). The swarm's advantage is in *budget*, not asymptote: it operates at this level with no reward signal and no training, and leads the learners until they overtake it, which takes ≈4,000 of the 15,000 training environment-steps (they plateau by $\approx 10^4$). (An earlier weak online baseline appeared to lose to the swarm outright; properly trained, it does not—we report the corrected result.) We scope this to one minimal setting—it *bounds* rather than settles the comparison—and read it not as beating reinforcement learning but as a strong *zero-shot, reward-free* operating point, worth pursuing where a reward signal or a training budget is scarce. The result is a concrete demonstration that the self-organizing physics of a minimal swarm can carry a collective close to the optimum reinforcement learning seeks, with none of reinforcement learning's machinery. The companion paper (Wilkerson, 2026) develops the underlying mechanism.

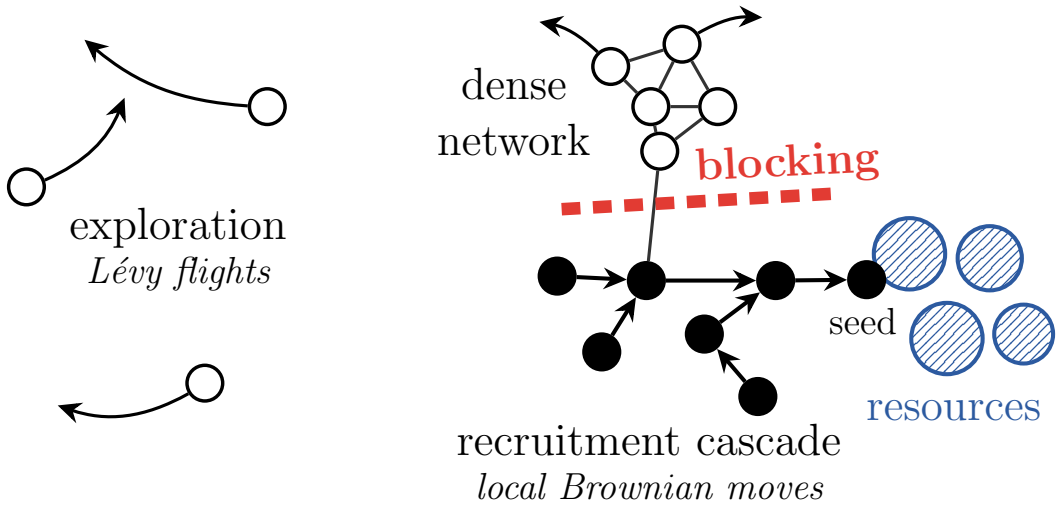


Figure 1: **The physical model.** Agents explore by Lévy flights (left). An agent that lands on a resource—a *seed*—triggers a recruitment cascade along the proximity network, drawing neighbours into local convergence on the find; as the neighbourhood crowds, degree-normalized activation a/k falls below threshold ϕ and further recruitment is *blocked* (red). Schematic of the model described in §3; the full model is developed in (Wilkerson, 2026).

1 Introduction

Reinforcement learning (RL) is the machine-learning account of the explore/exploit trade-off: a system must *explore* to discover where reward is and *exploit* what it has found, and good behaviour is the behaviour that balances the two and settles at the best available payoff (Sutton & Barto, 2018). A foraging swarm faces exactly this problem. Patches of resource are options of unknown location and yield; the swarm must search for them and converge on the good ones, with only local sensing and no central controller. Because many foragers share each patch, a forager's best available payoff is congestion-limited—the most it can get given where everyone else is—so the problem is multi-agent and its optimum is a collective *allocation* (each forager at its best available payoff: the ideal-free / Nash equilibrium), not a single best option. The natural question is whether a swarm that forages well is, in effect, reaching the optimum of this reinforcement-learning problem, and if so, what plays the part of the learner.

We answer: the swarm approaches the optimum—it attains the optimal allocation's *shape*—and there is no learner, and, more strongly, no reward signal inside the dynamics at all. A minimal self-organizing mechanism—Lévy exploration, threshold recruitment to discoveries, and density-dependent dispersal—balances exploration and exploitation and converges toward the optimal allocation of foragers across patches (reaching its shape, though it remains measurably exploitable; §4, §6). The point of this paper is a claim about *level of abstraction*. The reinforcement learning is at the level of the *problem solved* and the *outcome reached*, not at the level of the micro-rule. We are explicit about this because a tempting reading—that the recruitment rule is itself a reward-following learning update—is false, and we show it is false. The agents never evaluate how good a patch is; recruitment responds to local crowding, not to payoff. What is remarkable is precisely that a system with *none* of reinforcement learning's machinery—no value function, no policy parameters, no training, no reward signal—nonetheless lands near the optimum reinforcement learning is built to find.

Claims. The paper makes four claims, each stated so it can be checked against evidence.

1. **(The swarm approaches the explore/exploit optimum.)** The self-organizing cascade cycle balances exploration and exploitation and converges to the *shape* of the optimal foraging

allocation—the ideal free distribution (matching slope 0.94)—reached dynamically and caused by the recruitment–movement coupling (remove it and matching collapses to 0.08); the exact equilibrium is not reached—the swarm stays measurably exploitable (§4, §6).

2. **(Learner-free and reward-free.)** It does so with no value function, no policy parameters, and no training—and the dynamics contain no reward signal: the agents never read a payoff, and recruitment is driven by crowding, not by reward (§5).
3. **(It reaches the allocation, it does not learn.)** Reaching this allocation is not epistemic learning: the configuration's informativeness rises once and plateaus, and re-settles after the resources move with no cumulative gain (§5).
4. **(A strong zero-shot operating point.)** Matched for embodiment—same environment, sensing, movement, and actions—the reward-free swarm reaches ≈65% of a validated oracle ceiling at zero training; properly-trained PPO/CTDE learners converge modestly *above* it (≈74–75%) but only after ≈4,000 environment steps of training (of 15,000), so the swarm leads the learner through the low-budget / cold-start regime, at no training, reward signal, or parameters (§6). It does not, however, transfer to a different reward structure—a fixed, reward-blind mechanism cannot.

What is, and is not, claimed. We claim that the *optimum* of a reinforcement-learning problem—the optimal explore/exploit allocation, the ideal-free / Nash equilibrium—is reached by a physical, self-organizing mechanism with no learner. We do *not* claim that the micro-dynamics are a reinforcement-learning algorithm, a reward-following update, or the replicator equation: the recruitment rule is reward-free *by construction* (no step reads payoff; §3, §5). We do *not* claim the optimum is a total-reward maximum—under resource sharing it is an equal-intake equilibrium (App. A). And we do *not* claim epistemic learning: the swarm reaches the optimum, it does not improve an estimate from experience. Return genuinely varies—and the explore/exploit stakes bite—only in the *dynamic, embodied* setting (§6), where cumulative intake differs across policies; that is where we compare to a learner. The content is both precise and, we think, important: a system with *none* of reinforcement learning's machinery lands exactly where learning-in- games dynamics converge, and in the embodied setting is a strong zero-shot operating point—it leads a trained learner through the low-budget regime, though a well-trained learner converges modestly above it (§6). How the comparison comes out depends sharply on the learner's training budget, a caution for how such comparisons are run.

Scope and the companion paper. The swarm's mechanism—a spatial threshold recruitment cascade whose self-organized "cascade window" is analysed in full—is developed in a companion paper (Wilkerson, 2026), which we cite for the complete treatment. This paper is *self-contained*: every claim it makes is supported by evidence shown here. In particular, the self-organizing operating regime is demonstrated in-paper (§4, Fig. 5), the recruit-state movement property we use is a stated feature of the model (below), and the learner comparison matches movement by construction (App. D)—so no load-bearing claim depends on the companion. We summarise the mechanism only as far as the present reading needs; our claims concern the explore/exploit problem and the allocation the swarm reaches.

2 Related work

The explore/exploit problem and bandits. The multi-armed bandit (Robbins, 1952; Lattimore & Szepesvári, 2020) is the canonical formalisation of explore/exploit, and the one-state reduction of the general RL problem (Sutton & Barto, 2018). Our setting is its multi-agent, resource-sharing instance: arms are patches whose per-capita yield falls as more foragers work them, so the optimum is an allocation, not a single best arm.

The optimum: ideal free distribution and habitat games. The optimal allocation of foragers over patches is the *ideal free distribution* (IFD): foragers distribute in proportion to resource so that per-capita intake is equal and no forager gains by moving (Fretwell & Lucas, 1969). This is the Nash equilibrium of the habitat-selection (congestion) game (Rosenthal, 1973; Wardrop, 1952) and the standard optimality criterion in behavioural ecology. We use the IFD only to *characterise the solution* the swarm reaches; we do not claim the swarm reaches it by playing the game or by any game-solving procedure.

Learning in games as the mean field of RL. Reward-proportional reinforcement is, in the large-population mean field, the replicator equation (Börgers & Sarin, 1997), the bridge used to analyse multi-agent learning dynamically (Hofbauer & Sigmund, 1998; Sandholm, 2010; Tuyls & Nowé, 2005; Bloembergen et al., 2015). This is the level at which one *might* hope to identify a physical swarm with RL. We deliberately do *not* rest our claim there: §5 shows the swarm's micro-dynamics are reward-free and are not a replicator update. Our identification is at the problem/solution level instead.

Swarm optimisation, stigmergy, and designed game-solvers. Metaheuristics such as particle-swarm (Kennedy & Eberhart, 1995) and ant-colony optimisation (Dorigo et al., 1996) coordinate many simple units, but they are *designed* to optimise a supplied objective and carry memory (a personal/global best; a pheromone table). Stigmergic foraging is usually read as *reinforcement*, with the pheromone field an external memory / value estimate (Vellinger et al., 2025). Decentralised multi-agent control reaches Nash / potential-game equilibria from local information *by design* (Marden et al., 2009), and bee-inspired allocation reaches a Nash, optimal allocation by designed local rules (Quijano & Passino, 2010). Ours is none of these: it is not a designed optimiser, it carries no memory or value estimate, and it uses no reward—it is a physical mechanism whose self-organized dynamics happen to settle at the problem's optimum. That contrast makes the claim precise: the swarm is the memory-free, reward-free limit these methods are not.

Minimally intelligent agents reaching an optimum. That non-optimising agents can reach a market's competitive allocation is the "zero-intelligence" result of experimental economics (Gode & Sunder, 1993). Our result is in this spirit—the structure of the interaction, not the sophistication of the agent, does the optimising—extended to a spatial, self-organizing foraging swarm and to the explore/exploit problem. Reaching an equilibrium without computing payoffs also connects to no-regret and evolutionary learning dynamics, which provably converge to equilibria of the underlying game (Sandholm, 2010; Hofbauer & Sigmund, 1998); our mechanism is the limiting case that uses no regret signal and no payoff at all. We make no entropy-regularised ("soft") RL claim (Haarnoja et al., 2018; Ziebart et al., 2008): the spread of the converged allocation is the ideal free distribution set by resource sharing, not a temperature.

3 The physical model: a self-organizing forager

The full model, its critical "cascade window," and the efficiency analysis are in the companion paper (Wilkerson, 2026); here is what the present reading needs, drawn in Figs. 1 and 2. N foragers move on a torus. An inactive agent *explores* by heavy-tailed (Lévy) hops. An agent that lands on resource becomes *active* (a *seed*) and *recruits* its neighbours: recruitment is a linear-threshold cascade on the proximity graph (an edge joins agents within sensing range r). An agent activates when the active fraction of its neighbours crosses a threshold, $a/k \geq \phi$ (a active neighbours, k neighbours), and recruited agents switch to short local moves toward the find. Because activation is *degree-normalized*, the same rule that spreads a cascade also stifles it once a neighbourhood crowds and a/k falls back below ϕ: recruitment to a crowded patch shuts off. So the swarm aggregates on a find and then releases to explore again. Crucially for what follows, **no step of this rule reads a payoff**: a seed fires on *contact* with resource, and recruitment responds to the *active fraction* of neighbours—a crowd signal—never to how rich a patch is or how much intake an agent receives.

The cascade cycle is an explore/exploit balancer. The mechanism runs a drive–avalanche cycle (Fig. 2): discovery seeds a cascade (explore → recruit → exploit), crowding stifles and disperses it (exploit → released → explore), and the swarm self-organizes to a poised operating point—a "cascade window" in connectivity—without any parameter being tuned. We show this self-tuned operating regime directly in this paper (§4, Fig. 5): the swarm holds a subcritical, near-ideal-matching operating point across a 60× density range with no per-condition tuning. The window's lower edge is the percolation onset and its upper edge the degree-dependent blocking bound; the companion paper maps it in full with phase diagrams over connectivity and activation threshold (Wilkerson, 2026). The density-dependent dispersal ("blocking") is thus what regulates the explore/exploit *balance*—how much of the swarm explores versus exploits. At any instant part of the swarm is exploring and part is exploiting; the cycle continually re-apportions the two—the explore/exploit trade-off managed, in real time, by the physics rather than by a controller. One further property of the model is used below: recruitment is *movement-neutral*—by definition of the recruit state,

Symbol	Meaning	Value
N	foragers	200
L	torus side length	800
$\rho_r = N(r/L)^2$	connectivity density	2.1 ($r \approx 82$)
ϕ	activation threshold ($a/k \geq \phi$)	0.18 ($2/\phi \approx 11$)
α	Lévy exponent	1.0
C	patches (resource clumps)	10
β	Pareto patch-richness exponent	2.2
M	resource sources	80
z_c^{RGG}	percolation onset	$\approx$4.51

Table 1: **Main model parameters** (spatial foraging model, §3). The abstract congestion game (§6, App. C) uses K=10 patches and N=200 agents with the same Pareto richness; the embodied comparison (App. D) uses N=200 over horizon T=600.

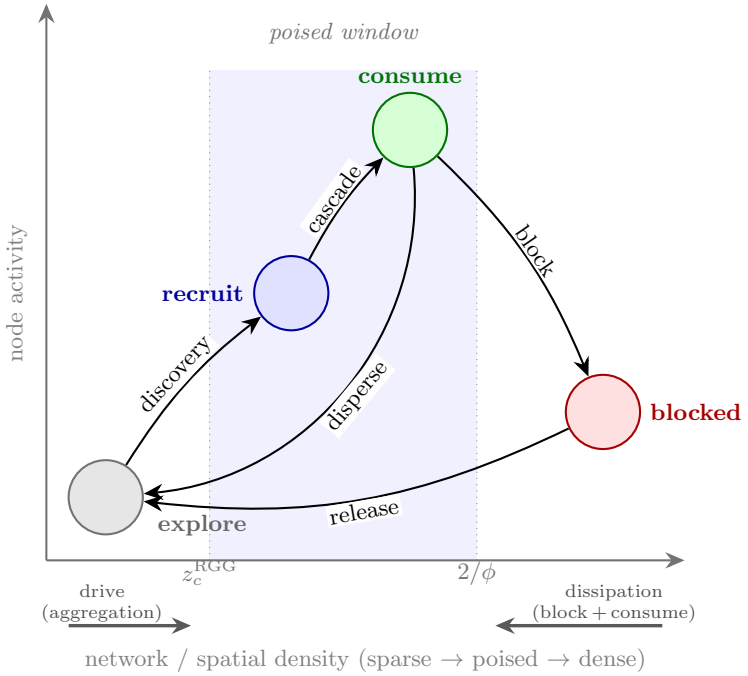


Figure 2: **The self-organizing cycle balances exploration and exploitation.** In the plane of network/spatial density (horizontal) and node activity (vertical), the swarm runs a drive–avalanche cycle: discovery seeds a recruitment cascade (explore→recruit→exploit); crowding blocks and disperses it (exploit→released→explore); aggregation (*drive*) versus dispersal (*dissipation*) hold the swarm in a poised window, with no parameter tuned. Schematic of the self-organizing cycle described in §3; full treatment in (Wilkerson, 2026).

recruited agents reuse the exploring step length and change only direction, so no movement budget is spent to coordinate (a stated feature of the model here; analysed in (Wilkerson, 2026)).

4 The swarm approaches the explore/exploit optimum

The problem and its optimum. Coarse-grain the swarm by the fraction n_i of agents committed to patch i (the rest exploring); $\mathbf{n}$ is the swarm's allocation over patches. With resource Q_i shared among the Nn_i foragers working patch i, per-capita intake is $f_i = Q_i/(Nn_i)$, which falls as a patch crowds. The optimal allocation—where no forager can raise its intake by moving—equalises per-capita intake across

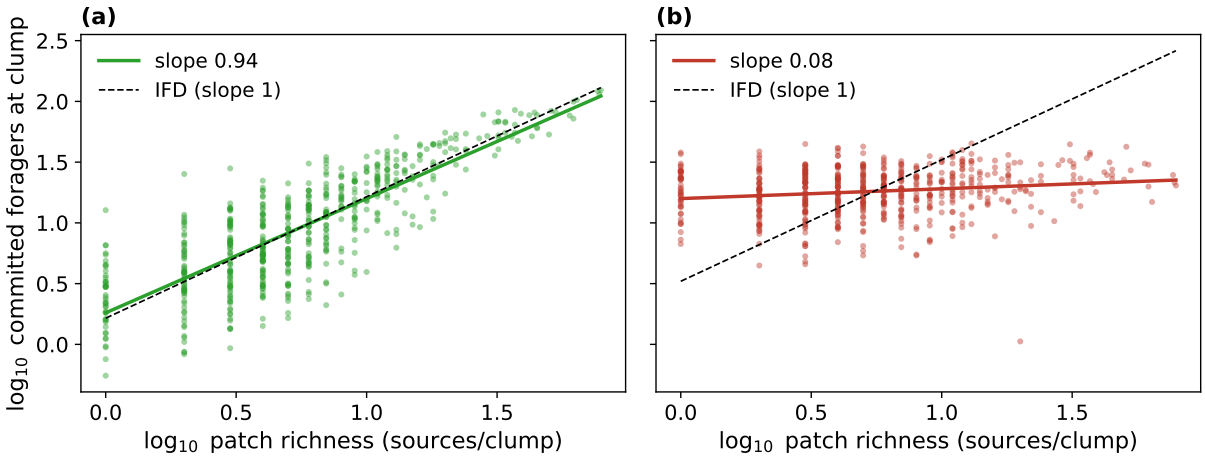


Figure 3: **The swarm reaches the optimal allocation, and the coupling causes it.** Committed foragers per patch vs. richness, pooled over B=64 independent replica worlds (n=591 patch-replica points). **(a) Recruiting.** Pooled slope 0.94 (R^2=0.67; per-replica mean slope ≈1.0); dashed is the ideal slope 1, the small shortfall the "undermatching" real foragers show. **(b) Coupling off.** With recruitment removed, matching is destroyed (slope 0.08): the optimal (ideal-free) allocation is caused by the recruitment–movement coupling.

occupied patches, giving $n_i \propto Q_i$: the ideal free distribution, and the Nash equilibrium of the habitat-sharing game (Fretwell & Lucas, 1969; Rosenthal, 1973) (App. A). This is the optimum of the swarm's multi-agent explore/exploit problem: the allocation at which every forager is at its best available payoff. Because resource is shared, total intake $\sum_i n_i f_i = \sum_i Q_i/N$ is the same for any interior allocation (App. A), so this optimum is a load-balancing *equilibrium*, not a total-harvest maximum; the harvesting-efficiency side of foraging—that finding and exploiting resources quickly collects more over time—is the companion paper's result (Wilkerson, 2026).

The swarm reaches it, and the coupling causes it. If the swarm settles at the ideal free distribution, committed foragers per patch should track patch richness with unit slope. In a heterogeneous field (Pareto patch richness over a ∼40-fold range—heavy-tailed, sparse, and clustered, as natural resource fields typically are) the measured slope is 0.94 ($R^2 = 0.67$; Fig. 3a)—near-ideal, with the mild *undermatching* real foragers universally show. This is the *shape* of the optimum (foragers in proportion to richness); how far the swarm sits from the *exact* equilibrium—its exploitability—is a separate question we quantify against a trained learner in §6, where it is measurably exploitable, not at the Nash point. The decisive result for the causal claim is the ablation: *removing the recruitment coupling collapses the slope to* 0.08 (Fig. 3b). The allocation shape is not a property of the individual agents—it is produced by coupling recruitment to movement. (We attribute no finer mechanism to the coupling than this: in particular we do *not* claim the coupling implements a particular learning update—§5.)

The allocation is reached dynamically. The matching is not imposed; it is *reached*. Tracking the matching slope through a run (Fig. 4), the coupled swarm climbs from near zero to its ideal-free-shape plateau within ∼150 steps and holds; the uncoupled swarm stays flat. The allocation is an *attractor* of the explore/exploit dynamics—reached quickly and then maintained.

The operating regime is self-organized, not tuned. The swarm is not placed at a hand-chosen operating point. Sweeping connectivity ρ_r over a 60× density range (Fig. 5), the swarm holds a *subcritical* branching ratio σ nearly flat (0.06–0.09)—with its (broad) minimum near the operating point we use—and maintains near-ideal matching throughout (slope 0.80–1.03), with no per-condition tuning. We are careful about the sense of "critical." The single-seed recruitment cascade has the classic window between the

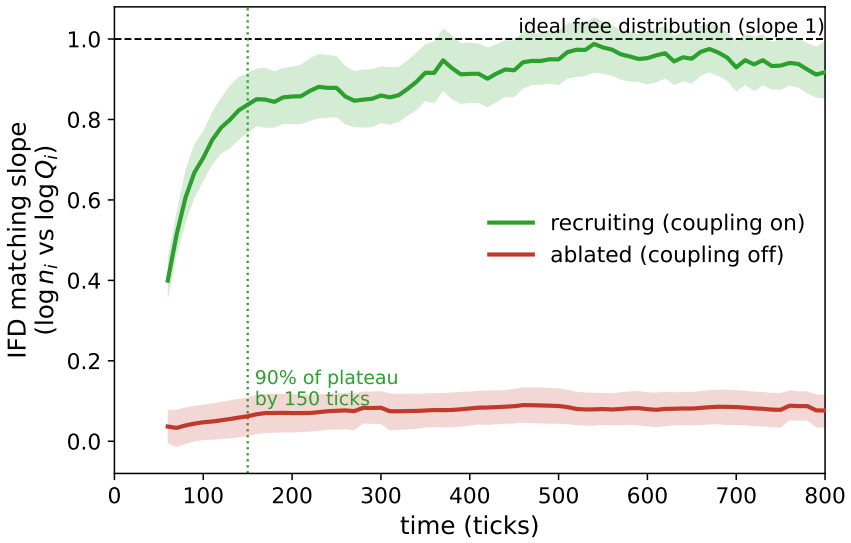


Figure 4: **The allocation is reached dynamically.** Matching slope (regression of log committed foragers per patch on log richness, trailing window; replicas with bootstrap 95% CI) vs. time. With the recruitment coupling (green) the swarm climbs to the ideal-free plateau (slope $\to 1$, dashed) within $\sim$150 steps and holds; without it (red) matching never develops.

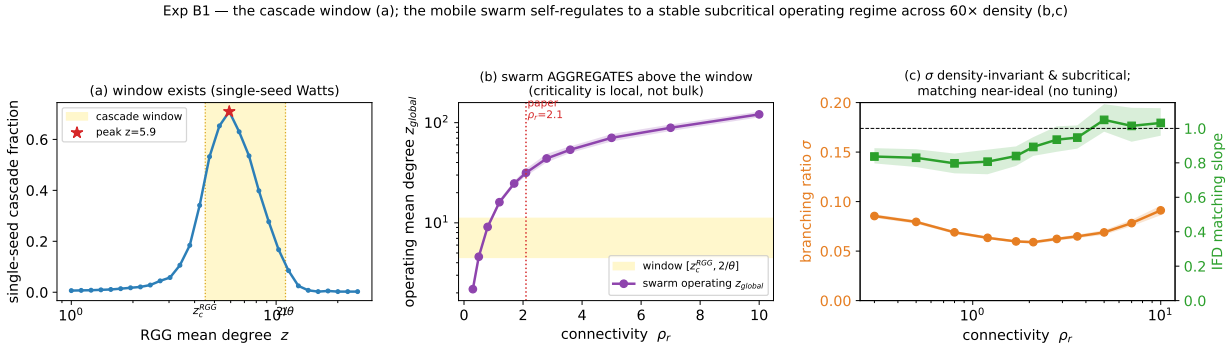


Figure 5: **The operating regime is self-organized, not tuned. (a)** The single-seed recruitment cascade has an inverted-U window between $z_c^{\mathrm{RGG}} \approx 4.51$ and the blocking bound $2/\phi$ (reproduced here). **(b)** Swept over a 60× connectivity range, the mobile swarm holds a subcritical branching ratio σ nearly flat (0.06–0.09), with its broad minimum near the operating point used throughout ($\rho_r = 2.1$); its bulk mean degree runs *above* the single-seed window (aggregation), so the criticality is local, not bulk. **(c)** Near-ideal IFD matching (slope 0.80–1.03) holds across the whole range with no per-condition tuning.

percolation onset $z_c^{\mathrm{RGG}} \approx 4.51$ and the blocking bound $2/\phi$ (Fig. 5a); but because recruited agents aggregate, the swarm's *bulk* mean degree runs *above* that window, so the criticality is *local* (clusters near-critical, global branching subcritical), as the companion paper develops (Wilkerson, 2026). What is self-organized here is the *regime*—a density-invariant subcritical operating point and robust near-ideal matching, reached without tuning—not a bulk mean degree sitting inside the cascade window.

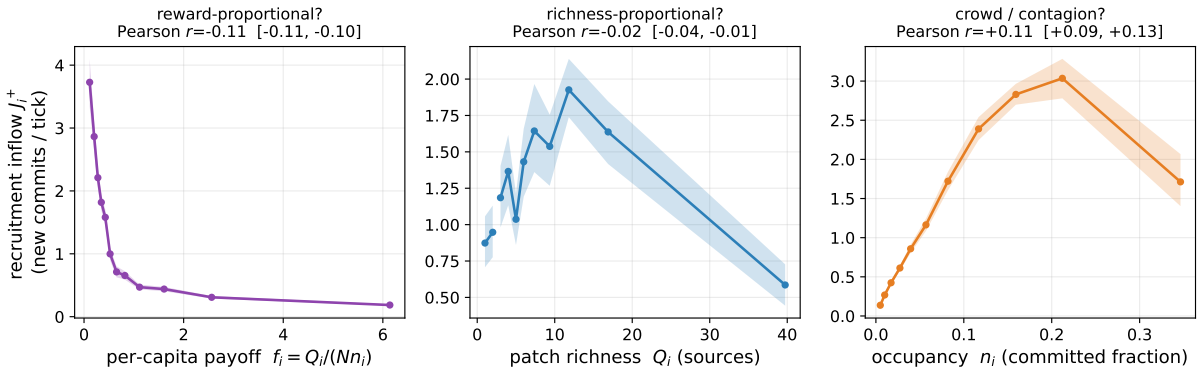


Figure 6: **Illustrative: recruitment tracks crowd, not payoff** (reward-freeness itself is established by the rule's construction, §3, not by this figure). Per-patch recruitment inflow J_i^+ (new commits per step) vs. per-capita payoff f_i, richness Q_i, and occupancy n_i (quantile-binned mean, bootstrap 95% CI over 64 replicas). Inflow correlates positively with occupancy ($r = +0.11$)—the crowd signal the cascade rule uses—weakly negatively with payoff ($r = -0.11$), and shows no monotone trend with richness (non-monotone, linear $r \approx 0$). The correlations are small and, under equal sharing, non-identifying (App. B); they illustrate rather than prove the reward-free construction.

5 How it reaches the optimum: learner-free, reward-free, and not by learning

The swarm reaches the optimum. This section establishes *how it does not*: without a learner, without a reward signal, and without learning over time. These absences are the content of "learner-free," and we show each directly rather than asserting it.

No learner. There are no adjustable parameters and no training phase. Ordinary RL (Sutton & Barto, 2018) keeps a value function or a policy network and updates it over many episodes; the trained parameters are the product. Here the instantaneous configuration $\mathbf{n}(t)$—carried by the agents' positions and their interaction network, stored nowhere else—*is* the allocation, taken in real time, with nothing trained or stored.

No reward signal: recruitment is reward-free by construction. The decisive evidence is the rule itself: per-capita intake f_i is never computed or read by any agent—a seed fires on contact with resource, and recruitment responds to the active *fraction* of neighbours (§3). There is simply no reward in the loop, so no reward can be followed; f_i is a quantity *we* compute to describe the outcome, not a signal the dynamics use. Measurements of the flow are consistent with this but, as we note below and in App. B, cannot by themselves establish it. Illustratively (Fig. 6), logging per-patch recruitment inflow over 64 replicas, inflow correlates *positively* with local occupancy n_i (Pearson $r = +0.11$)—the crowd signal the cascade rule uses—weakly *negatively* with per-capita payoff f_i ($r = -0.11$), and shows no monotone trend with richness Q_i (non-monotone, linear $r \approx 0$). The negative payoff relation is mechanical: at a patch $n_i f_i = Q_i/N$, so a patch recruitment has crowded necessarily has low per-capita intake. These correlations are small and illustrative, not an independent proof of reward-blindness—under equal sharing payoff-rank and distance-from-equilibrium coincide (App. B), so no correlational test can separate "reward-free" from "reward-following." The proof is the construction. The upshot is that the optimum is reached by a mechanism *blind to reward*: the agents pile onto richer patches only because richer patches, having more resource sites, are discovered and re-seeded more often, not because anyone measures that they pay better—the strongest sense of "without a learner," with no reward in the loop to learn from.

What the coarse dynamics are (and are not). Since the flow carries no reward, is it nonetheless a reward-following law in disguise? No. Fitting the denoised mean flow of the allocation over the transient to competing mean-field laws, by leave-replicas-out cross-validation (App. B), a reward-proportional *replicator*

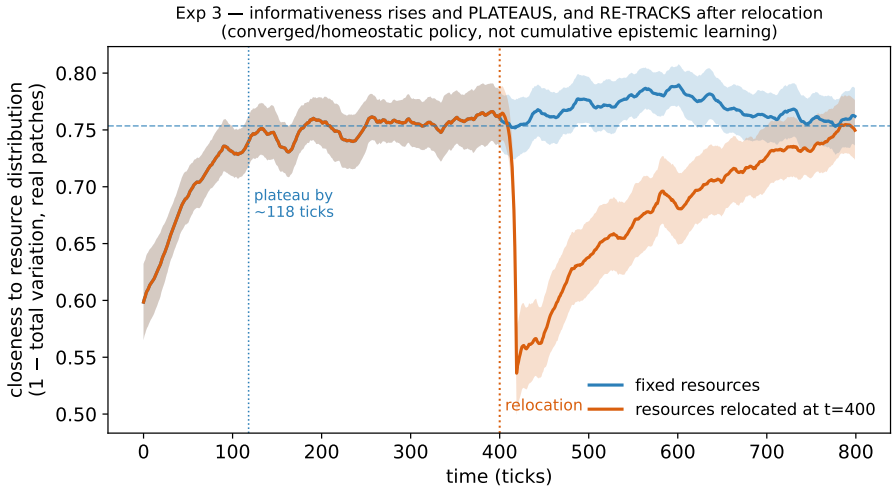


Figure 7: **The swarm reaches the optimum but does not learn it.** Allocation informativeness (closeness = 1− total-variation distance between committed occupancy and the resource distribution; replicas with bootstrap 95% CI) vs. time. It rises once and *plateaus* by ∼120 steps; after the resources are relocated (t=400, dashed line) it dips and *re-tracks* to the same plateau, not a higher one—no cumulative improvement. This is a converged policy, not a learner.

update is *not* the best description (held-out $R^2 = 0.26$); the flow is a *nonlinear relaxation* toward the ideal-free fixed point (held-out R^2 up to 0.64 for saturating forms). The coarse dynamics are thus a relaxation of the allocation to the optimum, not a reward-gradient ascent. We report this as a characterisation of the flow, not as an RL identification: the relaxation reaches the optimum; it is not a learning update.

It reaches the optimum, it does not learn. Reaching the optimum is not epistemic learning. The primary reason is architectural: there is no stored or accumulating state—no value estimate, no counts, no memory—so there is nothing in which experience could accumulate. Behaviourally this shows as two things (Fig. 7): under fixed conditions the allocation's informativeness about the resource field rises once and *plateaus* by ∼120 steps rather than improving with more experience; and when the resources are relocated mid-run, the swarm re-settles to the *same* quality it held before, with no cumulative gain (relocation at t=400). (The informativeness metric is bounded, so a plateau alone is only suggestive; the load-bearing evidence is the absent accumulating state plus the re-tracking to the same, not a higher, level.) The swarm reaches and *holds* the optimum, re-finding it when the world changes—a converged, homeostatic policy, not an agent accumulating knowledge.

6 Comparison to a trained learner: zero-shot operating point vs. converged learner

How does a learner-free mechanism compare to an actual learner? We compare in *two* settings. First, an *abstract* (disembodied) version of the congestion game—direct patch choice with a per-step payoff oracle and no spatial search—with independent softmax policy-gradient and Q-learning agents (App. C). Second, the *same embodied environment* the swarm acts in: one spatial world with identical local sensing, movement, and action set {explore, approach, stay}, where the *only* difference between arms is fixed (recruitment) versus learned control (App. D).

Disembodied, the learner wins on precision. On the abstract game the swarm reaches the right allocation *shape* at zero training (matching slope 0.95—consistent with the 0.94 in the spatial field, §4; 83% of the way to Nash by 1− total-variation distance; Fig. 8a) but is *not* near-optimal on exploitability (Nash-gap 0.58, roughly 50× a converged Q-learner's ≈ 0.011 and orders of magnitude above a policy-gradient learner's $\approx 5 \times 10^{-4}$; Fig. 8b). So a trained learner, handed a per-step payoff oracle, buys precision the

swarm lacks—and, as the next experiment shows, given enough budget it keeps that edge once embodied too. What the swarm has instead is a strong operating point at *zero* budget.

Embodied, the swarm is a strong zero-shot operating point. In the shared embodied environment we normalise by a *validated oracle ceiling*: the maximum total intake achievable under free placement given the current resource state, a strict upper bound (no realised run approaches it). The reward-free swarm reaches ≈65% of this ceiling at *zero* training and with *no reward signal* (Fig. 9), roughly 1.8× a random controller (≈36%). We then train two standard learners in the identical environment—independent PPO (IPPO: clipped surrogate, GAE, entropy; a shared actor with a per-agent local critic) and a centralised-critic CTDE learner (MAPPO: the same actor with a critic over privileged global state, decentralised execution)—each given a rich 14-dimensional local observation and a dense shaped reward, with evaluation always on the *true* intake fraction of the ceiling. The swarm stays reward-free; the learner is advantaged on *information*, not handicapped. Given adequate budget, both learners *exceed* the swarm: over 16 seeds each, IPPO reaches 0.75 (median 0.73) and MAPPO 0.74 (median 0.72) of the ceiling, and *every one of the* 32 *learner seeds finishes above the swarm mean* (Fig. 9b,c). The learning curves plateau—flat from ∼8,000 to 15,000 steps—so this is converged performance, not a snapshot mid-climb.

The advantage is in budget, not asymptote. What separates the two—and is the point—is *budget*, visible in the learning curves (Fig. 9a). The swarm sits at ≈65% of the ceiling from the first step, with no reward and no training. The learners start well below it and only cross the swarm after ≈4,000 environment steps—about a quarter of the 15,000-step training run—plateauing by $\approx 10^4$ steps. So the swarm *leads* the entire low-budget / cold-start regime up to that crossover and still lands within ≈0.10 of the converged learners (its best seeds, 0.75, reach into the learners' lower tail). An earlier version of this comparison appeared to show the swarm beating an embodied learner outright; that was a weak-baseline artifact—an under-trained online REINFORCE controller plateaus near 0.37, barely above the random 0.36, i.e. it essentially does not learn, and a properly-trained learner does not share that ceiling. The honest reading is narrower and, we think, more useful: the swarm's value is a strong *zero-shot, reward-free* operating point—no reward signal, no training, no parameters—that is competitive in the low-budget regime, not asymptotic superiority over reinforcement learning. We read this not as beating reinforcement learning but as evidence the mechanism is worth pursuing where a reward signal or a training budget is scarce. This is one minimal shared environment, which *bounds* rather than settles the general comparison.

What a learner still buys: transfer. One advantage of learning is unambiguous. In the abstract game's non-congestion regime (fixed per-patch reward, optimum = concentrate on the richest patch) the learners adapt (80–97% of optimum) while the reward-blind swarm keeps spreading and is sub-optimal (38%; Fig. 8c). A learner, by reading reward, transfers across reward structures; the fixed, reward-blind swarm cannot. (In the embodied environment this regime is not a clean adaptation test—there intake accrues per agent-on-source, so spreading is near-optimal and the swarm again leads—so we rest the transfer point on the abstract game.)

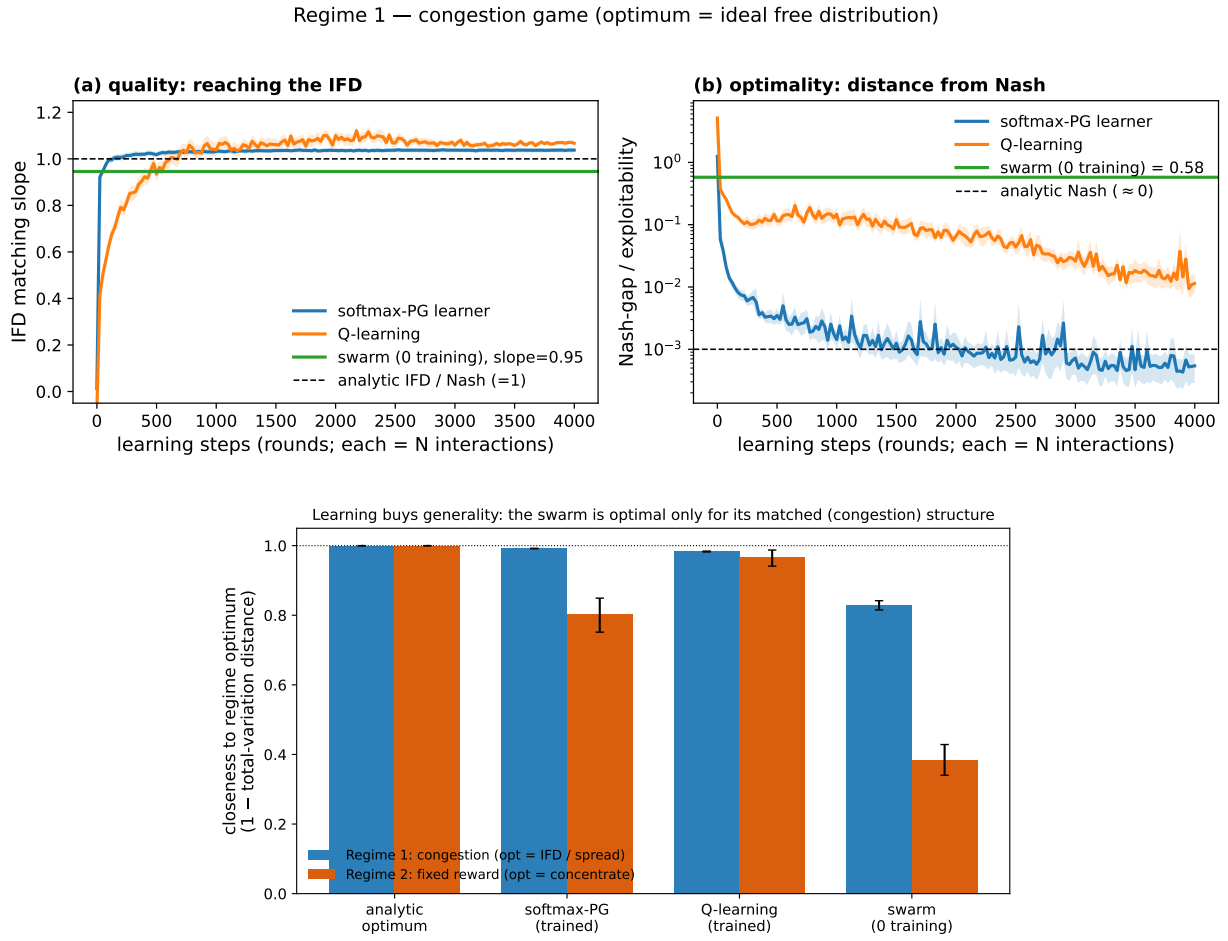


Figure 8: **Learner-free swarm vs. trained multi-agent RL, same optimum (abstract congestion game). (a)** matching slope and **(b)** Nash-gap (exploitability) vs. learning steps for independent softmax policy-gradient and Q-learning (bootstrap 95% CI over 64 replicas), with the zero-training swarm and the analytic optimum as references: the swarm reaches the right allocation *shape* at zero training (a), but its Nash-gap (0.58) stays well above the trained learners' (b)—it is not near-optimal on exploitability. The learners play an easier, disembodied problem (no spatial search); units differ (learner steps vs. swarm physical steps), as marked. **(c)** Closeness to the regime optimum in the *congestion* regime (optimum = ideal free distribution) and a *non-congestion* regime (fixed per-patch reward, optimum = richest patch). The swarm is near-optimal only in the congestion regime its mechanism matches; the learners adapt to both. Learning buys generality.

To reach the optimum	**Learner-free swarm (this work)**	**Trained multi-agent RL**
Reward signal	none (reward-free, Fig. 6)	per-agent scalar payoff, every step
Stored parameters	none	a policy (and critic) network
Training before use	none; acts at its operating point	≈4,000 env-steps to match, ≈10^4 to plateau (of 15,000)
Communication	local recruitment only (within range)	none, but needs a reward signal every step
Movement to coordinate	none beyond exploration (recruits reuse the exploring step)	learned from local observations
Precision, *abstract* game	right shape (slope 0.95), but exploitable (Nash-gap 0.58)	near-exact (Nash-gap ≈0)
Oracle-ceiling fraction, *embodied*	≈65% at *zero* training (Fig. 9)	≈74–75% converged; below the swarm until ≈4,000 steps
Low-budget / cold-start regime	leads (zero-shot)	below the swarm until it has trained
Transfer across reward structures	none (congestion/IFD only)	adapts

Table 2: **The two pay for different things.** The learner-free swarm reaches the allocation shape with no reward signal, no parameters, and no training, and is competitive in the low-budget regime; the trained

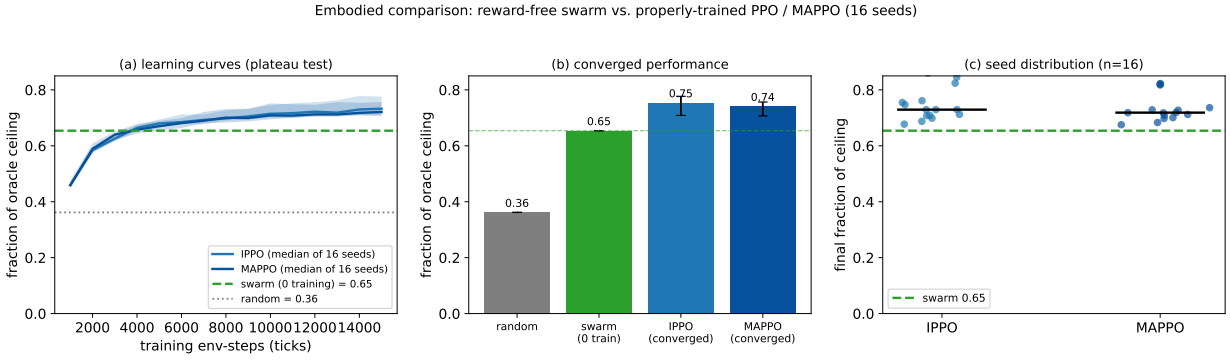


Figure 9: **Embodied comparison to properly-trained learners: a budget separation, not an asymptotic one.** Same spatial environment and matched sensing/movement/action set for all arms; fraction of a validated oracle ceiling; 16 seeds per learner. **(a)** learning curves (median and inter-quartile band) for independent PPO (IPPO) and a centralised-critic CTDE learner (MAPPO): both start below the reward-free swarm (dashed, zero training) and random (dotted), cross the swarm at ≈4,000 environment steps (of 15,000 total), and plateau at 0.72–0.75 by $\approx 10^4$ steps (flat tail ⇒ convergence demonstrated, not assumed). **(b)** converged fractions: random 0.36, swarm 0.65 (zero training, no reward), IPPO 0.75, MAPPO 0.74. **(c)** per-seed finals: all 32 learner seeds finish above the swarm mean; the swarm's best seeds (0.75) reach into the learners' lower tail. The swarm's value is the zero-shot, reward-free operating point—it leads the low-budget regime and lands within ≈0.10 of the converged learners—not asymptotic superiority.

7 Discussion

Where the reinforcement learning is. The contribution is a demonstration that the *optimum* of a reinforcement-learning problem—the explore/exploit optimum, here the optimal spatial allocation—can be *approached* by a self-organizing physical mechanism with none of reinforcement learning's machinery: the swarm attains the optimum's allocation *shape*, though not its exact equilibrium (it stays measurably exploitable, §6). The identification is at the level of the problem and the allocation reached, not the micro-rule: the dynamics evaluate no reward (by construction) and are not a reward-following update, and the system reaches this allocation without learning. Read this way, "learner-free reinforcement learning" is a careful statement, not a relabelling: the problem is reinforcement learning's, the outcome is (approximately) its optimum, and the learner is absent.

Why this is more than aggregation. A sceptic may ask whether "reaching the ideal free distribution" is just agents piling onto richer patches. Three things make it a genuine optimum rather than a trivial pile-up. First, the allocation it reaches has the *shape* of the optimum (foragers in proportion to richness, approaching equal per-capita intake), not an arbitrary aggregation—and the ablation shows it is non-trivial to reach (without recruitment the swarm does *not* distribute proportionally, slope 0.08). Second, the explore/exploit trade-off is actively and continually balanced by the cascade cycle (§3), not set once. Third, the optimum is an attractor: the swarm re-finds it when the resources move (Fig. 7). What it is *not* is a learner—and we are explicit about that boundary.

Limitations. (i) The optimum reached is the ideal free distribution of a resource-sharing (congestion) problem; the swarm's mechanism is matched to that structure and does not transfer to others (§6). That structure, however—sparse, patchily clustered resources (heavy-tailed in size and spacing) shared among foragers—is the common case in natural foraging environments, not a special one, so the domain the mechanism is matched to is broad and ecologically typical rather than contrived. (ii) Two forces set the allocation, and we separate them. The density-dependent dispersal ("blocking") regulates the explore/exploit *balance* and the critical operating point—shown here by the self-tuned subcritical regime across a 60× density range (Fig. 5), and mapped in full across connectivity and threshold in the companion paper (Wilkerson, 2026). The allocation *shape* (slope-one matching), by contrast, is set by richness-proportional discovery and recruit-

ment, not by the crowding cutoff: a single-operating-point ablation here finds the matching slope essentially unchanged when the cutoff is removed ($0.94 \to 0.92$; App. B), a consistency check alongside Part I's swept characterisation. (iii) The trained-learner comparison includes both an abstract game and an embodied, partially-observed contest (§6); in the embodied setting the swarm is a zero-shot operating point that leads in the low-budget regime, while properly-trained PPO/CTDE learners converge modestly above it (Fig. 9), so the claim is a *budget* separation, not asymptotic superiority. It is one minimal environment and two learner families, so it *bounds* rather than settles the general swarm-vs-RL question—a full state-of-the-art MARL pipeline is out of scope. (iv) We give no closed-form regret or convergence-rate bound.

Outlook. The embodied comparison is a budget crossover: the swarm leads from zero training and trained PPO/CTDE learners overtake it after ≈4,000 steps (Fig. 9). The natural next steps are to characterise *where* that crossover sits—how it moves with environment difficulty, nonstationarity, agent count, and learner architecture—and to give a convergence-rate / regret characterisation of the physical flow against standard bandit and MARL algorithms, including whether a hybrid that warm-starts a learner from the swarm's zero-shot allocation shortens its budget.

Why it matters. As an engineering primitive, the mechanism delivers a close approximation of the explore/exploit optimum—a load-balanced, near-ideal-free allocation—with no model, no reward signal, no training, and no fitted parameters, deployable on minimal agents. Matched for embodiment it is a strong zero-shot operating point (≈65% of an oracle ceiling) that leads a trained learner through the low-budget / cold-start regime, before a well-trained learner—given a reward signal and a budget—converges modestly above it (§6). It is a case where the optimisation *is* the physics of a self-organizing collective, not an algorithm run on top of it.

8 Conclusion

A foraging swarm faces the explore/exploit problem that defines reinforcement learning. We have shown that a minimal self-organizing swarm *approaches its optimum*—balancing exploration and exploitation through its cascade cycle and converging to the *shape* of the optimal allocation of foragers over patches (the ideal free distribution)—with no learner, no trained parameters, and no reward signal in its dynamics, and that it does so by *reaching* this allocation rather than by *learning* it. Given a reward signal and a training budget, a learner reaches the optimum more precisely, converges modestly higher in the embodied setting, and transfers to other reward structures; the swarm's return is a strong zero-shot operating point at none of that cost, leading in the low-budget regime—the price and the point of having no learner. The reinforcement learning here is the problem and the (approximate) optimum reached, emergent from physics.

Competing interests

The author has filed a provisional patent application related to the recruitment-switching mechanism described in this work.

Data and code availability

Complete simulation code, precomputed sweep data, figure-regenerating scripts, and an interactive browser-based demo of the mechanism are available from the author on request, and will be released publicly (with a hosted, runnable demo) upon publication.

A The optimal allocation: ideal free distribution and habitat-game Nash

With resource Q_i shared among the Nn_i foragers on patch i, per-capita intake is $f_i = Q_i/(Nn_i)$. An allocation is stable against unilateral relocation when per-capita intake is equal across all occupied patches, $f_i = \bar{f}$ for all occupied i. With $f_i = Q_i/(Nn_i)$ this gives $n_i \propto Q_i$: the ideal free distribution (Fretwell & Lucas,

1969). Because each agent's payoff depends only on how many share its patch, the habitat-selection game is an anonymous congestion game, whose pure Nash equilibrium is exactly this equal-intake allocation (Rosenthal, 1973; Wardrop, 1952); player-specific congestion games need not share this structure (Milchtaich, 1996). Note that mean per-capita intake $\langle f \rangle = \sum_i n_i f_i = \sum_i Q_i/N$ is constant on the interior, so the optimum is an equal-intake *equilibrium* (a proportional-fair, log-utility allocation (Kelly et al., 1998)), not a total-intake maximum: the swarm load-balances, it does not increase total intake. We use the ideal free distribution purely to *characterise* the solution the swarm reaches (slope-one matching, $n_i \propto Q_i$); the swarm does not compute payoffs or solve the game (§5).

B What the coarse dynamics are: model competition and the role of blocking

Model competition. We ask which mean-field law best describes the measured flow of the allocation $\dot{n}_i$. Per-step $\dot{n}_i$ is dominated by discrete-event noise (all candidate laws sit at held-out $R^2 \approx 0.03$ at step resolution), so we fit the *denoised mean flow*: samples binned by (richness stratum × occupancy), each model fit to the per-bin mean $\dot{n}_i$ over the transient (steps 1–150), with model selection by leave-16-replicas-out cross-validation and AIC/BIC. Candidates: a reward-proportional *replicator* $\dot{n}_i = \lambda\, n_i(f_i - \bar{f})$ (which on equal sharing is algebraically a *linear relaxation* to the fixed point, $\lambda'(n_i^* - n_i)$, $n_i^* = Q_i/\sum_j Q_j$); a saturating *best-response*/logit $\dot{n}_i = \kappa(\sigma_\beta(f)_i - n_i)$; a payoff-blind *logistic/contagion* $\dot{n}_i = \alpha\, n_i(cQ_i - n_i)$; and a constant null. Held-out R^2: logit 0.64, contagion 0.47, replicator/linear-relaxation 0.26, null 0. Two honest conclusions follow. (i) The linear replicator/relaxation form is *rejected* as sufficient—the flow has saturating curvature. (ii) The best-fitting forms are nonlinear *relaxations* toward the $n_i \propto Q_i$ fixed point; because on equal sharing payoff-rank equals gap-rank, the logit fit does not indicate that the swarm computes payoffs (it cannot—§5). The flow is a nonlinear relaxation of the allocation to the optimum, consistent with a reward-free mechanism. Two caveats keep this honest. First, under equal sharing the replicator regressor $n_i(f_i - \bar{f})$ equals the signed distance to the fixed point, $(Q_{\text{tot}}/N)(n_i^* - n_i)$ (measured correlation 0.997), so payoff-rank and gap-rank coincide and *no* regression on this data can separate a reward-free from a reward-following rule—reward-freeness rests on the rule's construction (§3), not on these fits. Second, the payoff-blind logistic/contagion form, though competitive in bulk R^2, is itself rejected by a shape diagnostic: at small occupancy the measured flow grows with the gap rather than being throttled toward zero as $n_i(cQ_i - n_i)$ would require. So the fits identify "nonlinear relaxation to the fixed point" but neither a specific reward-following law nor a specific payoff-blind one.

The role of density-dependent dispersal (blocking). We separate blocking's two possible roles: setting the explore/exploit *balance* versus setting the allocation *shape*. For the balance, blocking is causal: we show in-paper that the swarm self-tunes to a subcritical operating regime across a 60× density range with no per-condition tuning (§4, Fig. 5), and the companion paper maps the bounding cascade window in full with phase diagrams over connectivity and activation threshold (percolation onset below, degree-dependent blocking above) (Wilkerson, 2026). For the allocation shape, we run a single intermediate ablation here: removing the crowding cutoff (replacing the degree-normalized rule $a/k \geq \phi$ with an absolute-count rule, the maximal no-cutoff limit) while leaving recruitment otherwise intact. The matching slope is essentially unchanged ($0.94 \to 0.92$, no over-aggregation), while the active fraction rises ($0.73 \to 0.85$)—so blocking was holding more of the swarm exploring (it regulates the balance) but was not setting the proportional allocation. That shape traces to richness-proportional discovery and recruitment (richer patches, with more resource sites, are seeded more often). This is a single-operating-point consistency check, not a sweep (the self-tuning across conditions is Fig. 5; the full parameter map is in (Wilkerson, 2026)), and the absolute-count toggle is an extreme limit—but together they support attributing the allocation shape to the recruitment–movement coupling and discovery, while the self-organization (blocking) does the explore/exploit balancing.

C Trained multi-agent RL baseline

On an abstract version of the congestion allocation game (K patches, N agents, per-capita payoff $f_i = Q_i/(Nn_i)$, richness Q_i matched to the swarm's), each agent runs an independent learner with local, own-payoff information only: a softmax policy-gradient (REINFORCE with a per-agent baseline) or an ϵ-greedy

Q-learner. Quality is the matching slope and the distance to the ideal free distribution; the "regret" analogue is the *exploitability* / Nash-gap, since reward-regret is ill-defined here (mean per-capita intake is constant, App. A). The swarm, being reward-blind, is run at its physical operating point and its steady-state allocation compared on the same (Q_i, N, K). The non-congestion regime replaces the payoff with a fixed per-patch reward $f_i = Q_i$ (optimum = concentrate on $\arg\max_i Q_i$), where the swarm's fixed mechanism is sub-optimal while the learners adapt. Learner hyperparameters are lightly tuned so the baseline is neither crippled nor over-fit; the swarm has no tunable parameters, an asymmetry we state rather than hide.

D Embodied learner comparison

All arms share one spatial environment: a torus with $N = 200$ agents per replica world, a local sensing radius, Pareto-clumped resources that deplete and respawn, and the action set {explore (a Lévy/Brownian step), approach the nearest sensed signal, stay}. The *swarm* arm applies the fixed recruitment+blocking rule over this observation/action set, reward-free. Each *learner* arm replaces the rule with a trained policy over the same action set; only the controller differs—sensing, movement, and actions are identical—so the comparison isolates fixed versus learned control under matched embodiment and partial observability. Because both arms use the *same* movement primitives and step budget, the fairness of this comparison does not rely on the companion paper's movement-neutrality claim: it is matched by construction here.

Learners. We train two standard algorithms. *IPPO* is independent PPO—a clipped surrogate with generalised advantage estimation and an entropy bonus, a shared-weight actor with a per-agent *local* critic. *MAPPO* uses the same actor with a *centralised* critic over a 32-dimensional privileged global state (per-clump occupancy, richness, and live capacity, plus totals), with decentralised execution (CTDE). Both see a rich 14-dimensional egocentric observation (on-resource flag; nearest-resource proximity and direction; local and wide resource density; neighbour-foraging fraction and local degree; nearest on-food neighbour proximity and direction; own short-memory intake and on-food fraction)—at least as informative as what the swarm effectively uses, so the learner is advantaged on *information*, not handicapped. Training uses a dense shaped reward (per-step per-capita intake plus 0.05× resource-proximity); *evaluation always uses the true intake fraction of the oracle ceiling*, never the shaped reward. The oracle ceiling is $\min(N, \sum \text{cap})$ per step, a strict upper bound that no realised run (swarm, learner, or random) approaches. Hyperparameters (shared, lightly chosen on confirmation runs): learning rate 3×10^{-4}, a 64×2 tanh MLP, rollout 128, 4 epochs, 8 minibatches, clip 0.2, $\gamma = 0.99$, $\lambda = 0.95$, value coefficient 0.5, entropy 0.01, gradient-norm clip 0.5, Adam.

Protocol and result. Each learner is trained for 15,000 environment steps and scored on a trailing 2,000-step window; we run 16 seeds per algorithm (the seed is the unit of uncertainty, since the shared-weight policy couples replicas) and compare against multi-seed reward-free swarm and random-action references on the same fields. Fractions of the oracle ceiling: random 0.36; swarm 0.65 at zero training; IPPO 0.75 (median 0.73); MAPPO 0.74 (median 0.72). All 32 learner seeds finish above the swarm mean; the learning curves plateau (flat from ∼8,000 to 15,000 steps), and both learners cross the swarm at ≈4,000 environment steps—$\approx 6.4 \times 10^6$ agent-steps (4,000 steps × N=200 across the R=8 parallel replica worlds), about a quarter of the 15,000-step run (Fig. 9). Correctness checks: the learners rise far above the random 0.36 (they are clearly learning), and the same PPO implementation climbs the ideal-free matching slope on the easy abstract congestion game (App. C), confirming the implementation is sound. A note on an earlier version: a deliberately simple online REINFORCE controller plateaued near 0.37—barely above random—and an initial read mistook that weak baseline for evidence that an embodied learner cannot match the swarm. The properly-trained learners here correct that: the swarm's advantage is a zero-shot / low-budget one, not an asymptotic ceiling on the learner.

References

Daan Bloembergen, Karl Tuyls, Daniel Hennes, and Michael Kaisers. Evolutionary dynamics of multi-agent learning: A survey. *Journal of Artificial Intelligence Research*, 53:659–697, 2015.

Tilman Börgers and Rajiv Sarin. Learning through reinforcement and replicator dynamics. *Journal of Economic Theory*, 77(1):1–14, 1997.

Marco Dorigo, Vittorio Maniezzo, and Alberto Colorni. Ant system: optimization by a colony of cooperating agents. *IEEE Transactions on Systems, Man, and Cybernetics, Part B (Cybernetics)*, 26(1):29–41, 1996.

Stephen D Fretwell and Henry L Lucas. On territorial behavior and other factors influencing habitat distribution in birds. i. theoretical development. *Acta Biotheoretica*, 19:16–36, 1969.

Dhananjay K. Gode and Shyam Sunder. Allocative efficiency of markets with zero-intelligence traders: Market as a partial substitute for individual rationality. *Journal of Political Economy*, 101(1):119–137, 1993.

Tuomas Haarnoja, Aurick Zhou, Pieter Abbeel, and Sergey Levine. Soft actor-critic: off-policy maximum entropy deep reinforcement learning with a stochastic actor. In *International Conference on Machine Learning*, pp. 1861–1870. PMLR, 2018.

Josef Hofbauer and Karl Sigmund. *Evolutionary Games and Population Dynamics*. Cambridge University Press, 1998.

Frank P. Kelly, Aman K. Maulloo, and David K. H. Tan. Rate control for communication networks: shadow prices, proportional fairness and stability. *Journal of the Operational Research Society*, 49(3):237–252, 1998.

James Kennedy and Russell Eberhart. Particle swarm optimization. In *Proceedings of ICNN'95 – International Conference on Neural Networks*, volume 4, pp. 1942–1948. IEEE, 1995.

Tor Lattimore and Csaba Szepesvári. *Bandit Algorithms*. Cambridge University Press, 2020.

Jason R. Marden, Gürdal Arslan, and Jeff S. Shamma. Cooperative control and potential games. *IEEE Transactions on Systems, Man, and Cybernetics, Part B (Cybernetics)*, 39(6):1393–1407, 2009.

Igal Milchtaich. Congestion games with player-specific payoff functions. *Games and Economic Behavior*, 13 (1):111–124, 1996.

Nicanor Quijano and Kevin M. Passino. Honey bee social foraging algorithms for resource allocation: Theory and application. *Engineering Applications of Artificial Intelligence*, 23(6):845–861, 2010.

Herbert Robbins. Some aspects of the sequential design of experiments. *Bulletin of the American Mathematical Society*, 58(5):527–535, 1952.

Robert W. Rosenthal. A class of games possessing pure-strategy Nash equilibria. *International Journal of Game Theory*, 2(1):65–67, 1973.

William H. Sandholm. *Population Games and Evolutionary Dynamics*. MIT Press, 2010.

Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction*. MIT Press, Cambridge, MA, 2 edition, 2018.

Karl Tuyls and Ann Nowé. Evolutionary game theory and multi-agent reinforcement learning. *The Knowledge Engineering Review*, 20(1):63–90, 2005.

Vellinger, Antonic, and Tuci. From pheromones to policies: Reinforcement learning for engineered biological swarms, 2025. arXiv:2509.20095.

John Glen Wardrop. Some theoretical aspects of road traffic research. *Proceedings of the Institution of Civil Engineers*, 1(3):325–362, 1952.

Galen J. Wilkerson. A general mechanism for collective foraging: percolating recruitment cascades coupled to a maximum-entropy field, 2026. Companion paper (Part I), under review.

Brian D Ziebart, Andrew L Maas, J Andrew Bagnell, and Anind K Dey. Maximum entropy inverse reinforcement learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 8, pp. 1433–1438, 2008.